

Adiabatic switching for degenerate ground states

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● Plan of the presentation

- Motivation: Green's functions in many-body perturbation theory
- The Gell-Mann and Low formula in a simple case
- The degenerate case
- Back to Green's functions

● References:

- C. BROUDER, G. STOLTZ AND G. PANATI, Adiabatic approximation, Gell-Mann and Low theorem and degeneracies: A pedagogical example, *Phys. Rev. A* **72** (2008) 042102
- C. BROUDER, G. PANATI AND G. STOLTZ, Many-body Green function of degenerate systems, *Phys. Rev. Lett.* **103** (2009) 230401
- C. BROUDER, G. PANATI AND G. STOLTZ, Gell-Mann and Low formula for degenerate unperturbed states, *Ann. I. H. Poincaré-Phy* **10**(7) (2010) 1285-1309

Physical motivation

- The Gell-Mann and Low formula is a fundamental tool in Quantum Field Theory, used to obtain expressions of the Green's function amenable to a perturbative treatment
- The two point Green's function is defined as

$$G(t, r; t', r') = -i \left\langle \Psi_0, T[\psi_H(t, r) \psi_H^\dagger(t' x')] \Psi_0 \right\rangle$$

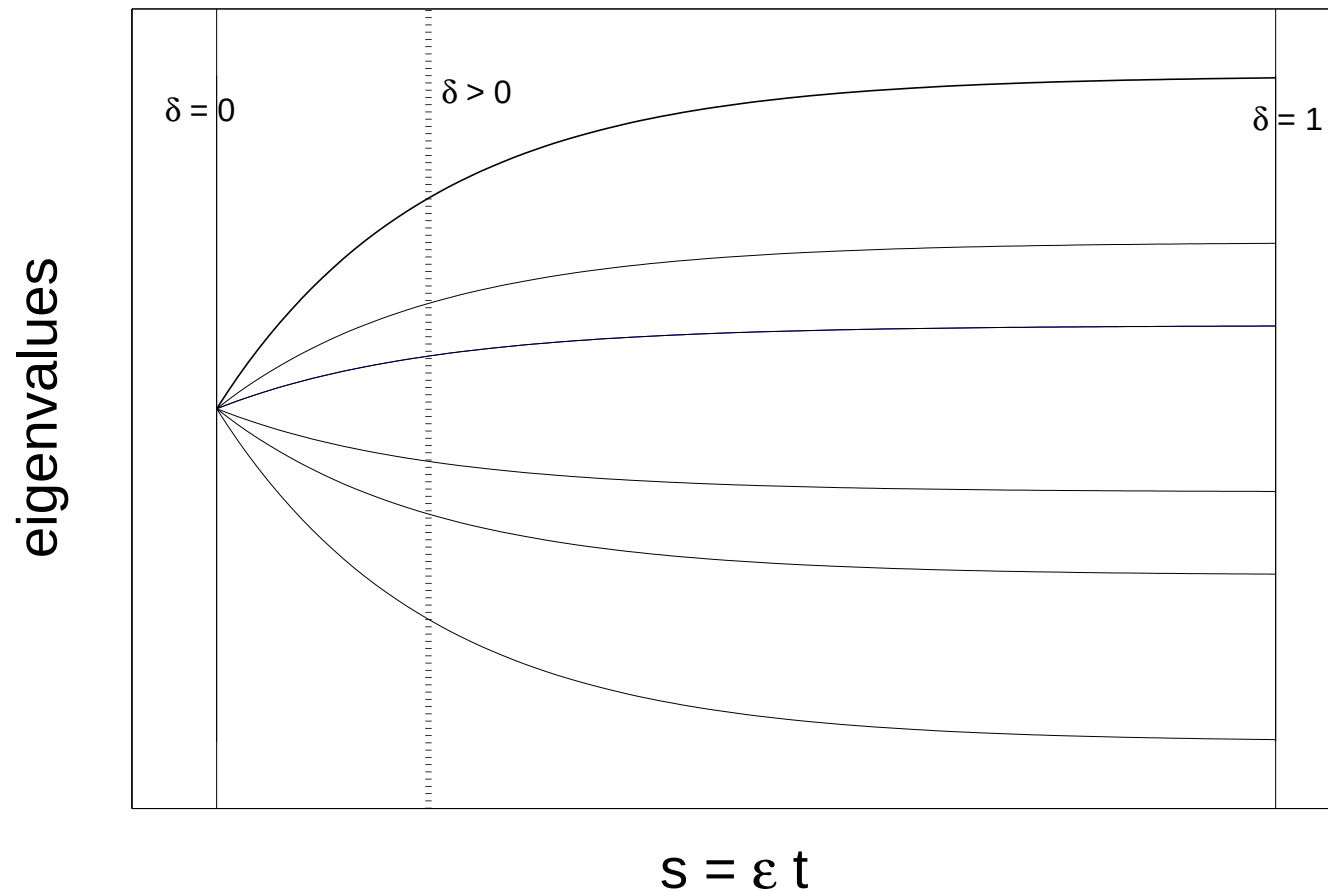
where Ψ_0 is the (unknown) ground state of some Hamiltonian $H_0 + V$

- When the ground state Φ_0 of H_0 is known, G can be expressed in terms of expectations with respect to Φ_0 using the Gell-Mann and Low formula:

$$G(t, r; t', r') = -i \lim_{\varepsilon \rightarrow 0} \frac{\left\langle \Psi_0, T[\psi_H(t, r) \psi_H^\dagger(t' x') U_\varepsilon(+\infty, -\infty)] \Psi_0 \right\rangle}{\left\langle \Psi_0, U_\varepsilon(+\infty, -\infty) \right\rangle}$$

- Formal expansions in terms of free-field Green's functions using Wick's theorem (Feynman diagrams)

The Gell-Mann and Low switching procedure requires some care when the ground state is degenerate... and this happens in many situations!



- Simplest possible system: Hamiltonian $H(t) = H_0 + e^{-\varepsilon|t|}H_1$ with

$$H_0 = \begin{pmatrix} \mu - \delta & 0 \\ 0 & \mu + \delta \end{pmatrix}, \quad H_1 = \begin{pmatrix} 0 & \alpha \\ \alpha & 0 \end{pmatrix}.$$

- Analytical computations can be performed
- The switching procedure is well defined when $\delta \neq 0$
- The switching procedure **fails for almost all initial states when $\delta = 0$** , and can be defined for **two specific states only!**

The Gell-Mann and Low formula in a simple case

Switching procedure (1)

- Consider, on a given Hilbert space $\mathcal{H}$,
 - a self-adjoint operator H_0 , with dense domain $D(H_0) \subset \mathcal{H}$
 - a **symmetric** perturbation V , H_0 -bounded with relative bound $a < 1$.
 - define $\tilde{H}(\lambda) = H_0 + \lambda V$ with $\lambda \in [0, 1]$
- **Switching function** $f \in C^2((-\infty, 0], [0, 1])$
 - non-decreasing
 - $f, f'' \in L^1((-\infty, 0])$
 - $f(0) = 1$ and $\lim_{\tau \rightarrow -\infty} f(\tau) = 0$
 - for $\tau \in (-\infty, 0]$, define $H(\tau) = \tilde{H}(f(\tau)) = H_0 + f(\tau) V$
- Denote by $U_\varepsilon(s, s_0)$ the unitary evolution generated by $H(\varepsilon s)$, i.e. the unique solution of the problem:

$$i \frac{dU_\varepsilon(s, s_0)}{ds} = H(\varepsilon s) U_\varepsilon(s, s_0), \quad U_\varepsilon(s_0, s_0) = \mathbb{I}$$

Switching procedure (2)

- **Divergent phase** as $\varepsilon \rightarrow 0$! Consider $V = 0$ and ϕ an eigenstate of H_0 :

$$U_\varepsilon(s, s_0) \phi = \exp \left(-\frac{iE_0(s - s_0)}{\varepsilon} \right) \phi$$

- Remove divergence by working in the **interaction picture**:

$$U_{\varepsilon, \text{int}}(s, s_0) = e^{isH_0} U_\varepsilon(s, s_0) e^{-is_0H_0}.$$

- **Macroscopic time** $t = \varepsilon s$: unitary evolution

$$i\varepsilon \frac{dU^\varepsilon(t, t_0)}{dt} = H(t) U^\varepsilon(t, t_0), \quad U^\varepsilon(t_0, t_0) = \mathbb{I},$$

so that, in the interaction picture, $U_{\text{int}}^\varepsilon(t, t_0) = e^{itH_0/\varepsilon} U^\varepsilon(t, t_0) e^{-it_0H_0/\varepsilon}$

- Standard results show that, for $\psi \in D(H_0)$, the following limit exists:

$$U_{\text{int}}^\varepsilon(t, -\infty) \psi = \lim_{t_0 \rightarrow -\infty} U_{\text{int}}^\varepsilon(t, t_0) \psi$$

- In order for eigenstates to be **stable** during the switching procedure, some **gap** conditions are required
- The spectrum of $\tilde{H}(\lambda) = H_0 + \lambda V$, $\lambda \in [0, 1]$, consists of **two disconnected pieces**

$$\sigma(\tilde{H}(\lambda)) = \sigma_N(\lambda) \cup \left(\sigma(\tilde{H}(\lambda)) \setminus \sigma_N(\lambda) \right)$$

where $\sigma_N(\lambda) = \left\{ \tilde{E}_j(\lambda), j = 1, \dots, N \right\} \subset \sigma_{\text{disc}}(\tilde{H}(\lambda))$

- There is **a uniform gap** between the two parts of the spectrum, and between the elements of $\sigma_N(\lambda)$, in the sense that:

$$\Delta(\lambda) = \min_{j=1, \dots, N} \left(\min \left\{ \left| \tilde{E}_j(\lambda) - E \right|, E \in \sigma(\tilde{H}(\lambda)) \setminus \{ \tilde{E}_1(\lambda), \dots, \tilde{E}_N(\lambda) \} \right\} \right),$$

$$\delta(\lambda) = \min \left\{ \left| \tilde{E}_j(\lambda) - \tilde{E}_i(\lambda) \right|, 1 \leq i < j \leq N \right\}$$

are bounded from below by a positive constant for all $\lambda \in [0, 1]$

- For simplicity, eigenvalues $E_j(\tau) = \tilde{E}_j(f(\tau))$ of multiplicity 1
- Then, for an eigenstate ψ_j of H_0 associated with $E_j(-\infty)$, if

$$\|P_j(-\infty) - P_j(0)\| < 1,$$

the limit

$$\Psi_j = \lim_{\varepsilon \rightarrow 0} \frac{U_{\text{int}}^\varepsilon(0, -\infty)\psi_j}{\langle \psi_j | U_{\text{int}}^\varepsilon(0, -\infty)\psi_j \rangle}$$

exists and is an eigenstate of $H_0 + V$ corresponding to the eigenvalue $E_j(0) = \tilde{E}_j(1)$

- First proof due to NENCIO and RASCHE (*Helvetica Physica Acta*, 1989)
- Extension to the case of eigenspaces of multiplicity higher than 1 provided some direction ϕ exists such that the denominator does not vanish...

First step of the proof: Geometric evolution

- **Kato intertwining** operator: $\frac{d\tilde{A}(\lambda, \lambda_0)}{d\lambda} = \tilde{K}(\lambda) \tilde{A}(\lambda, \lambda_0)$ with $\tilde{A}(\lambda_0, \lambda_0) = \mathbb{I}$
- Generator $\tilde{K}(\lambda) = - \sum_{j=1}^{N+1} \tilde{P}_j(\lambda) \frac{d\tilde{P}_j}{d\lambda}(\lambda)$, with $\tilde{P}_{N+1}(\lambda) = \mathbb{I} - \sum_{j=1}^N \tilde{P}_j(\lambda)$
- Since $\tilde{K}(\lambda)$ is uniformly bounded (gap, hence projectors smooth), the operator $\tilde{A}(\lambda, \lambda_0)$ is well-defined and strongly continuous
- $\tilde{A}(\lambda, \lambda_0)$ is **unitary** (since $K^* = -K$), and intertwines the spectral subspaces:

$$\tilde{P}_j(\lambda) = \tilde{A}(\lambda, \lambda_0) \tilde{P}_j(\lambda_0) \tilde{A}(\lambda, \lambda_0)^*$$

- Denoting by $A(s, s_0) = \tilde{A}(f(s), f(s_0))$,

$$P_j(0) A(0, -\infty) \psi_j = A(0, -\infty) P_j(-\infty) \psi_j = A(0, -\infty) \psi_j,$$

so that $A(0, -\infty) \psi_j$ is an **eigenstate** of $H(0) = H_0 + V$

Second step: Adiabatic evolution (adding the dynamical phase)

- **Adiabatic evolution** operator $U_A(s, s_0)$ is defined as the unique solution of

$$i \frac{dU_A(s, s_0)}{ds} = H_A(s) U_A(s, s_0), \quad U_A(s_0, s_0) = \mathbb{I},$$

where the **adiabatic Hamiltonian** is $H_A(s) = H(s) + iK(s)$

- U_A is also an **intertwiner**
- A and U_A **differ only by a phase**, which commutes with the spectral projectors: Define

$$\Phi(s, s_0) = A(s, s_0)^* U_A(s, s_0),$$

so that $U_A(s, s_0) = A(s, s_0) \Phi(s, s_0)$. Then, $[\Phi(s, s_0), P_j(s_0)] = 0$

- The **time-evolution of the phase** matrix is then easily obtained and

$$U_A(s, s_0) P_j(s_0) = \exp \left(-i \int_{s_0}^s E_j(r) dr \right) A(s, s_0) P_j(s_0)$$

Second step: Adiabatic evolution (rescaling the dynamical phase)

- Important again to work in the **interaction picture** to remove the divergent (dynamical) phase: $U_{A,\text{int}}(s, s_0) = e^{isH_0} U_A(s, s_0) e^{-is_0H_0}$

- It can be shown, through some limiting procedure, that

$$U_{A,\text{int}}(0, -\infty) P_j(-\infty) = \exp \left(-i \int_{-\infty}^0 E_j(r) - E_0 dr \right) A(0, -\infty) P_j(-\infty)$$

- Phase well-defined since $|E_j(r) - E_0| = |\tilde{E}_j(f(r)) - \tilde{E}_j(0)| \leq C f(r)$

- In the **time-rescaled** variable $t = \varepsilon s$,

$$U_{A,\text{int}}^\varepsilon(0, -\infty) P_j(-\infty) = \exp \left(-\frac{i}{\varepsilon} \int_{-\infty}^0 E_j(\tau) - E_0 d\tau \right) A(0, -\infty) P_j(-\infty).$$

- **Eliminate the phase** using

$$\frac{P_j(0) \psi_j}{\|P_j(0) \psi_j\|^2} = \frac{A(0, -\infty) \psi_j}{\langle \psi_j | A(0, -\infty) \psi_j \rangle} = \frac{U_{A,\text{int}}^\varepsilon(0, -\infty) \psi_j}{\left\langle \psi_j \left| U_{A,\text{int}}^\varepsilon(0, -\infty) \psi_j \right. \right\rangle},$$

Third step: Adiabatic limit of the full evolution

- **Compare** the adiabatic and full evolutions in the rescaled time-variable:

$$i\varepsilon \frac{dU_A^\varepsilon(t, t_0)}{dt} = \left(H(t) + i\varepsilon K(t) \right) U_A^\varepsilon(t, t_0), \quad i\varepsilon \frac{dU^\varepsilon(t, t_0)}{dt} = H(t) U^\varepsilon(t, t_0)$$

- Prove the **uniform convergence** $\lim_{\varepsilon \rightarrow 0} \|U^\varepsilon(0, -\infty) - U_A^\varepsilon(0, -\infty)\| = 0$
(although $U^\varepsilon(0, -\infty), U_A^\varepsilon(0, -\infty)$ do not have limits as $\varepsilon \rightarrow 0$)
- Strategy from (TEUFEL, *Adiabatic perturbation theory in quantum dynamics*, 2003):

$$U^\varepsilon(t, t_0) - U_A^\varepsilon(t, t_0) = -U^\varepsilon(t, t_0) \int_{t_0}^t U^\varepsilon(t_0, t') K(t') U_A^\varepsilon(t', t_0) dt'$$

- Define $\mathcal{K}(t) = -i\varepsilon U^\varepsilon(t_0, t) F(t) U^\varepsilon(t, t_0)$ with $[H(t), F(t)] = K(t)$. Then

$$\mathcal{K}'(t) = U^\varepsilon(t_0, t) [H(t), F(t)] U^\varepsilon(t, t_0) - i\varepsilon U^\varepsilon(t_0, t) F'(t) U^\varepsilon(t, t_0)$$

- **Similar to** $\int_0^t e^{-i\tau/\varepsilon} d\tau = i\varepsilon \left(e^{-it/\varepsilon} - 1 \right) = \text{highly oscillatory integral}$

Third step: Adiabatic limit of the full evolution (2)

- Expression of $F(t)$: useful to **keep track of the dependence on the gap** (required to understand the degenerate case)

$$F(t) = -\frac{1}{2} \left(\sum_{j=1}^{N+1} F_j(t) + G_j(t) \right), \quad F_j(t) = \frac{1}{2i\pi} \oint_{\Gamma_j(t)} P_j^\perp(t) R(z, t) \dot{R}(z, t) dz$$

where $R(z, t) = (H(t) - z)^{-1}$ and $\Gamma_j(t)$ is a contour enclosing $E_j(t)$ and no other element of the spectrum

- Similar definitions for G_j , F_{N+1} , G_{N+1}

- **Bounds** $\|F(t)\| \leq C_F \frac{f'(t)}{f(t)}$ and

$$\int_{t_0}^t \|F'\| \leq C \left(\frac{1}{f(t_0)} \int_{t_0}^t (|f''| + (f')^2) + \frac{1}{f(t_0)^2} \int_{t_0}^t (f')^2 \right)$$

The degenerate case

- Initial state is **degenerate**: $\tilde{E}_j(0) = \tilde{E}_k(0)$ for all $1 \leq j, k \leq N$
- Degeneracy splitting** (for simplicity): $\mathcal{P}_0 V \mathcal{P}_0$ has non-degenerate eigenvalues and for any $\lambda^* > 0$, there exists α such that

$$\inf_{\lambda^* \leq \lambda \leq 1} \min_{k \neq l} \left| \tilde{E}_k(\lambda) - \tilde{E}_l(\lambda) \right| \geq \alpha > 0$$

- Let $(\psi_1, \dots, \psi_N)$ be an basis of $\mathcal{E}_0$ which **diagonalizes the bounded operator $\mathcal{P}_0 V \mathcal{P}_0|_{\mathcal{E}_0}$** . Then, if $\|P_j(-\infty) - P_j(0)\| < 1$, the limit

$$\Psi_j = \lim_{\varepsilon \rightarrow 0} \frac{U_{\text{int}}^\varepsilon(0, -\infty) \psi_j}{\langle \psi_j | U_{\text{int}}^\varepsilon(0, -\infty) \psi_j \rangle}$$

exists and is an eigenstate of $H_0 + V$ corresponding to $E_j(0) = \tilde{E}_j(1)$

- Several extensions: decomposition to avoid the condition $\|P_j(-\infty) - P_j(0)\| < 1$; extension the case when $\mathcal{P}_0 V \mathcal{P}_0|_{\mathcal{E}_0}$ has degenerate eigenvalues; existence of finitely many eigenvalue crossings

Characterization of the initial states

- Theorem II.6.1 in (KATO, *Perturbation Theory for Linear Operators*) shows that the eigenvalues $\tilde{E}_j$ and projectors $\tilde{P}_j$ are **analytic** functions of λ
- Initial states defined from $P_j^{\text{init}} := \tilde{A}(0, \lambda) \tilde{P}_j(\lambda) \tilde{A}(\lambda, 0)$. **Characterization?**
- Eigenvectors satisfy $\tilde{H}(\lambda) \phi_j(\lambda) = \tilde{E}_j(\lambda) \phi_j(\lambda)$ with

$$\tilde{E}_j(\lambda) = \sum_{n=0}^{+\infty} \lambda^n E_{j,n}, \quad \phi_j(\lambda) = \sum_{n=0}^{+\infty} \lambda^n \varphi_{j,n}$$

- Hierarchy of equations. First order condition

$$(H_0 - E_0) \varphi_{j,1} = (E_{j,1} - V) \varphi_{j,0}$$

- A **necessary** condition for this equation to have a solution is that the right-hand side belongs to $\mathcal{E}_0^\perp$
- This requires $E_{j,1} = \langle \varphi_{j,0}, V \varphi_{j,0} \rangle$ and $\forall k \neq j, \langle \varphi_{k,0}, V \varphi_{j,0} \rangle = 0$ so that the basis **diagonalizes** $\mathcal{P}_0 V \mathcal{P}_0|_{\mathcal{E}_0}$

- Geometric and adiabatic evolutions: **unchanged** (the regularity of the projectors follows from the analytic continuation at $\lambda = 0$)
- Adiabatic limit: **decomposition** of the evolution into

$$U^\varepsilon(0, t_0) - U_A^\varepsilon(0, t_0) = -U^\varepsilon(0, t_0) \int_{t_0}^T U^\varepsilon(t_0, t) K(t) U_A^\varepsilon(t, t_0) dt \\ - U^\varepsilon(0, t_0) \int_T^0 U^\varepsilon(t_0, t) K(t) U_A^\varepsilon(t, t_0) dt$$

- an evolution on $[T, 0]$, for Hamiltonians operators with (small) **gaps** of order $f(T)$; bound in $C\varepsilon(1 + f(T)^{-2})$
- an evolution on the time-frame $(-\infty, T]$, with **T small enough** so that the unitary evolutions are not very different; bound in $Cf(T)$
- choose T such that $f(T) = \varepsilon^{1/3}$ to have a **final bound in $\varepsilon^{1/3}$**

Physical extensions

Application to Green's functions (formal)

- Operator A expressed in the **Heisenberg** picture $A_{\text{hsnbrg}}(t) = e^{itH} A e^{-itH}$ and, in the **interaction** picture, $A_{\text{int}}(t) = e^{itH_0} A e^{-itH_0}$
- Correlation function $C_{A,B}(t, t') = \langle \psi | T [A_{\text{hsnbrg}}(t) B_{\text{hsnbrg}}(t')] | \psi \rangle$
- Technical lemma: For fixed t, t' ,

$$\text{s-}\lim_{\varepsilon \rightarrow 0} U_{\varepsilon, \text{int}}(t, 0)^* A_{\text{int}}(t) U_{\varepsilon, \text{int}}(t, t') B_{\text{int}}(t') U_{\varepsilon, \text{int}}(t', 0) = A_{\text{hsnbrg}}(t) B_{\text{hsnbrg}}(t')$$

- Using the Gell-Mann and Low formula, it can then be shown that

$$C_{A,B}(t, t') = \lim_{\varepsilon \rightarrow 0} \frac{\langle \psi_0 | T [A_{\text{int}}(t) B_{\text{int}}(t') U_{\varepsilon, \text{int}}(+\infty, -\infty)] | \psi_0 \rangle}{\langle \psi_0 | U_{\varepsilon, \text{int}}(+\infty, -\infty) | \psi_0 \rangle}.$$

- Formal extension to the case when A, B are **field operators**
- Basis for a **perturbative treatment** of the Green's function, where the operators $U_{\varepsilon, \text{int}}(+\infty, -\infty)$ in the numerator and denominator are expanded using Feynman diagrams.